Opioid-Related ED Cohort Pipeline (opioid_ed)

Phase 0: Virginia APCD Longitudinal Lake (2016–2020)
Medical Claims Lake (All-Payer Encounters, Diagnosis & Procedure Positions 1–10)
Pharmacy Claims Lake (Adjudicated Dispensing, NDC Mappings, Days Supply)
Annual Partitions (2016, 2017, 2018, 2019) Stratified into 8 Age Bands (0–12 to 85–114)
Claims Stream
Enrollment Roster
Phase 1: Case Identification & Exclusivity (phase1_data_preparation.py)
10-Position Diagnostic Scanner Scans Primary through 10th ICD Position
Qualifying Emergency Encounter
Opioid ED Target Cases (t_0) • ICD-10 F11.20 (Opioid Dependence) • Acute Poisonings: T40.0X–T40.4X Timestamp: first_opioid_ed_date
Validate Exclusivity
Cohort Exclusivity Boundary Strictly drops patients meeting Polypharmacy ED criteria
1 Target Case
Validated Cases
Phase 3: 1:5 Matched Case–Control Sampling (phase3_cohort_creation.py)
Eligible Control Pool • Covered APCD members in same partition year • Zero opioid-related diagnoses in study year • Verified health system contact continuity
Candidate Pool
1:5 Incidence-Density Matching (No Replacement) Exact Stratification On: 1. Age Band (8 Strata: 0–12, 13–24, 25–44, ..., 85–114) 2. Biological Sex (Exact M/F Match) 3. Calendar Year Partition (2016–2019) 4. Active Coverage Continuity
Assign Anchor
Pseudo-Index Reference Date (t_pseudo) Controls assigned earliest non-ED medical claim date Guarantees identical 12-month baseline window
5 Matched Controls
Phase 2: Temporal Exposure & Leakage Audit (phase2_event_processing.py)
12-Month Pre-Index Baseline Window [t_0 - 365 days → t_0 - 1 day] • Longitudinal Opioid Titration & MME Escalation • Sedative Overlap (Benzodiazepines, Gabapentinoids) • CPIC CYP2D6/CYP2C19 Substrate Burden
Audit All Features
Step 3b Temporal Reverse-Causality Audit Purges 348 Post-Index Treatment Codes: • 164 OUD Meds (Buprenorphine, Naloxone, Vivitrol) • 84 Diagnostic CPTs (CPT 80307 Urine Drug Screens) • 100 Hospital Post-Stabilization Billing Codes
Lock Clean Features
Locked 682-Feature Whitelist Strictly derived from pre-index claims Applied symmetrically across Cases & Controls
Tabular Feature Matrix
Phase 4: QA Assertion & Checkpointed Export (phase4_finalization.py)
QA Passed
Cell-Size Suppression Guard Suppresses n ≤ 10 cell counts per CMS/VHI privacy policy
Write Gold Artifact
Final Parquet Fact Table (s3://pgx-repository/cohorts/) • schema: final_cohort_schema.json • Cases + 5:1 Controls with 682 Whitelisted Features • Checkpointed to S3 pgx-pipeline-status/

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DuckDB QA Unit Assertion (qa_index_date_uniqueness.py)
SELECT mi_person_key, count(DISTINCT first_opioid_ed_date)
HAVING count > 1;
Asserts EXACTLY ONE index date per person-year
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